Eric J. Pabon Cancel

Curriculum Vitae

Contact Information

Mathematical Sciences Building, Office 609
Department of Mathematics, Purdue University

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Research Interests

Dynamical Systems (e.g. Mathematical Physics, Symplectic Geometry, Modeling in Material Science), Machine Learning (e.g. Physics-Informed Neural Networks), Computational & Applied Mathematics (e.g. Numerical Methods, Mathematical Algorithms, Data Assimilation).

Education & Background

Doctor of Philosophy in Mathematics — May 2029
Purdue University, West Lafayette, Indiana | Advisor: Prof. Aaron Yip

Master of Science in Mathematics — May 2026
Purdue University, West Lafayette, Indiana

Bachelor of Science in Pure Mathematics (*Magna Cum Laude*) — December 2022
Curricular Sequence in Applied Mathematics for Science and Engineering
University of Puerto Rico, Mayagüez Campus (UPRM), Mayagüez, Puerto Rico | Mentor: Reyes M. Ortiz Albino

Skills and Other Information

Programming & Computation: Python, Julia, C++, SageMath, MATLAB | **Formatting & Tools:** HTML, LaTeX, Git
Math & AI/ML: NumPy, SciPy, PyTorch, TensorFlow, JAX, Flux, CUDA | **Spoken Languages:** English and Spanish

Research Experience

URA-Sandia Graduate Student Summer Fellow
URA-Sandia Graduate Student Summer Fellowship (administered by ORISE and ORAU) — May 2025–August 2025
Computational & Information Sciences Foundation, Sandia National Laboratories — Livermore, California
Supervised by: Dr. Moe Khalil & Dr. Rileigh Bandy, Sandia National Laboratories
Research Topics: Machine Learning-based Surrogate Models, Data Assimilation

GEM Fellow Summer Research Program Intern
MIT Lincoln Laboratory Summer Research Program (GEM Fellowship Employer Sponsor) — May 2023–August 2023
Group 39, Division 3, MIT Lincoln Laboratory, Massachusetts Institute of Technology — Lexington, Massachusetts
Supervised by: Dr. Sam Polk & Dr. Mabel Ramírez, MIT Lincoln Laboratory
Research Topics: Classical Machine Learning, Bayesian Inference, Generative AI, Unsupervised Learning

Research Opportunity for Undergraduate Students in STEM
Puerto Rico Louis Stokes Alliance for Minority Participation — August 2019–December 2022
Department of Mathematical Sciences, University of Puerto Rico, Mayagüez Campus — Mayagüez, Puerto Rico
Supervised by: Prof. Reyes M. Ortiz Albino, University of Puerto Rico at Mayagüez
Research Topics: Number Theory, Commutative Algebra, Factorization Theory in Integral Domains

NSF REU Participant

- Summer@ICERM 2022: Computational Combinatorics — June 2022–August 2022
  Institute for Computational and Experimental Research in Mathematics, Brown University — Providence, Rhode Island
  Supervised by: Prof. Pamela E. Harris, University of Wisconsin-Milwaukee
  Research Topics: Enumerative Combinatorics, Parking Functions
- NSF REU in Combinatorics, Probability and Algebraic Coding Theory — June 2021–August 2021
  East Tennessee State University & University of Puerto Rico at Ponce — Johnson City, Tennessee
  Supervised by: Prof. Fernando Piñero González, University of Puerto Rico at Ponce
  Research Topics: Algebraic Geometry & Linear Error Correcting Codes

Projects

Project in AI Topic (TBD)
URA STEM Ambassadors Program — May 2026–December 2026
2026 URA STEM Ambassadors Challenge
TBD

Project in Biotechnology

MIT Lincoln Laboratory Summer Research Program — June 2023–July 2023

2023 MIT Lincoln Laboratory Intern Innovative Idea Challenge (I^3C) | Mentors: Ryan Burrow and Ashok Kumar

SKINS: Skin-growth boosting and Intra-absorptive Solution bandages (3rd Place Project Winner)

Awards

Fellowships, Scholarships and Prizes

2025 Universities Research Association-Sandia National Labs Graduate Summer Fellowship — May 2025–August 2025

2023 National GEM Consortium PhD Science Fellowship — August 2023–May 2024

- Purdue University Department of Mathematics Sponsorship — August 2023–May 2024
- MIT Lincoln Laboratory Employer Sponsorship (Internship) — May 2023–August 2023

2023 MIT Lincoln Laboratory I^3C 3rd Place Research Proposal Prize — July 2023

2022 Evertec Inc. STEM Scholarship — October 2022

Puerto Rico-Louis Stokes Alliance for Minority Participation Research Scholarship — August 2019–December 2022

Merits and Recognitions

2026 URA STEM Ambassador — March 2026

2023 Ford Foundation Predoctoral Fellowship Honorable Mention — March 2023

2022 Hispanic Scholarship Fund Scholar — June 2022

National Math Alliance Predoctoral Scholar — November 2021

UPRM Faculty of Arts and Sciences Honor Roll — August 2018–May 2023

Puerto Rico Department of Education Math Olympiad, 2nd Place Regional Winner — February 2018

National Trig-Star Math Competition, 16th Overall Finalist — June 2017

Eagle Scout Rank, with 2 Silver Palms — May 2017

Puerto Rico Department of Education Math Olympiad, 1st Place District Winner — February 2016, 2017

Papers and Articles (The asterisk symbol (*) denotes alphabetical order authorship.)

Research Articles and Preprints:

[1] S. Polk, <u>E.J. Pabon-Cancel</u>, R. Paleja, K. Chestnut-Chang, R. Jensen and M. Ramirez.
Unsupervised Behavior Inference from Human Action Sequences (UNBIAS).
2024 IEEE Conference on Games (CoG), Milan, Italy, 2024, pp. 1-8.
DOI: https://doi.org/10.1109/CoG60054.2024.10645587

[2] *A. Allen, <u>E.J. Pabon-Cancel</u>, F. Piñero-Gonzalez and L. Polanco.
Improving the Dimension Bound of Hermitian-Lifted Codes.
arXiv: https://arxiv.org/abs/2302.01557

[3] *D. Chen, P.E. Harris, J. Carlos Martinez Mori, <u>E.J. Pabon-Cancel</u> and G. Sargent.
Permutation Invariant Parking Assortments.
Enumerative Combinatorics and Applications, **4:1**, 1-25 (2024). #S2R4.
DOI: https://doi.org/10.54550/ECA2024V4S1R4

[4] *I. Byrne, N. Dodson, R. Lynch, <u>E.J. Pabon-Cancel</u> and F. Piñero-Gonzalez.
Improving the minimum distance bound of Trace Goppa codes.
Designs, Codes and Cryptography. **91**, 2649–2663 (2023).
DOI: https://doi.org/10.1007/s10623-023-01216-6

Contributions to the profession:

[1] *P.E. Harris, Z. Markman, L. Martinez, A. Mock, <u>E.J. Pabón-Cancel</u>, A. Verga, and S. Wang.
A Model for a One-Hour Workshop on Mentoring.
MAA Focus, **43**(1), 18-21 (2023).

Service and Outreach

Academic Service

- Universities Research Association
  URA STEM Ambassadors Program — March 2026–present
  University Ambassador (Purdue University) — West Lafayette, Indiana

Teaching and Grading

- MA 26100: Multivariate Calculus (Asynchronous) — June 2026–August 2026

- MA 59500MB: Mathematical Biology (Grading) January 2026–May 2026
- MA 32500: History of Mathematics (Grading) January 2026–May 2026
- MA 26100 REC: Multivariate Calculus Recitation (Teaching) August 2025–December 2025
  January 2025–May 2025
  August 2024–December 2024
- MA 13900: Mathematics for Elementary Teachers III (Grading) June 2024–August 2024

Student Associations

- Purdue SACNAS Chapter
  Purdue University August 2025–May 2026
  Member West Lafayette, Indiana
- PythagoRUM
  University of Puerto Rico, Mayagüez Campus August 2022–December 2022
  Co-founder & Vice-President Mayagüez, Puerto Rico
- Society of Physics Students, UPRM Chapter
  University of Puerto Rico, Mayagüez Campus August 2018–December 2022
  Committee Assistant Mayagüez, Puerto Rico

Poster Sessions, Panels, Presentations and Conferences

- Purdue University Student Computational and Applied Mathematics Seminar 27 March 2026
  Helen B. Schleman Hall, Purdue University West Lafayette, Indiana
  Panel: Internships and Job Oppoortunities Beyond Academia
- Purdue Graduate Student Government Research-O-Rama 21 February 2026
  West Lafayette Public Library West Lafayette, Indiana
  Poster: Data-Driven Closure Models (DDCMs)
- Purdue University Prospective Student Visit Weekend Graduate Student Research Expo. 20 February 2026
  Mathematical Sciences Building, Purdue University West Lafayette, Indiana
  Poster: Data-Driven Closure Models (DDCMs)
- Purdue University Student Computational and Applied Mathematics Seminar 13 February 2026
  Helen B. Schleman Hall, Purdue University West Lafayette, Indiana
  Presentation: Data-Driven Closure Models (DDCMs)
- URA-Sandia Graduate Summer Fellowship Lightning Talk 6 August 2025
  Presentation: Data-Driven Closure Models (DDCMs) Virtual Seminar
- Sandia National Laboratories CA SIP Intern Symposium 5 August 2025
  Auditorium, Sandia National Laboratories-Livermore Livermore, California
  Poster: Data-Driven Closure Models (DDCMs)
- Combinatorics and Coding Theory in the Tropics (UPR-Ponce) 18 July 2025
  Invited REU Seminar: My Story & Permutation-Invariant Parking Assortments Virtual Seminar
- Purdue University Student Commutative Algebra Seminar 18 November 2024
  Helen B. Schleman Hall, Purdue University West Lafayette, Indiana
  Presentation: Results in $\tau_{(n)}$-factorizations and $\tau_{(n)}$-primes.
- Purdue University Student Math History Seminar 9 September 2024
  Lawson Computer Science Building, Purdue University West Lafayette, Indiana
  Presentation: Testimonios: Stories of Latinos and Hispanics in Mathematics
- Underrepresented Students in Topology and Algebra Research Symposium 2024 20-21 April 2024
  University of Iowa Iowa City, Iowa
- 2023 MIT Lincoln Lab Intern Innovative Idea Challenge 14, 21 July 2023
  MIT Lincoln Laboratory Auditorium Lexington, Massachusetts
  Poster: Skin-Absorptive and Skin-Growth Boosting Bandages
  Presentation: SKINS: Skin-growth boosting and Intra-absorptive Solution Bandages

- Combinatorics and Coding Theory in the Tropics (UPR-Ponce) — 7 July 2023
  Invited REU Seminar Talk: Graduate School: Application tips and advice — Virtual Seminar
- 2023 ACS Junior Technical Meeting-Puerto Rico Interdisciplinary Scientific Meeting — 29 April 2023
  University of Puerto Rico at Bayamón, Sponsored by PR-LSAMP — Bayamón, Puerto Rico
  Presentation: Properties of $\tau_{(n)}$-primes
- 38th Interuniversity Mathematical Sciences Research Seminar — 24-25 February 2023
  University of Puerto Rico, Mayagüez Campus — Mayagüez, Puerto Rico
  Presentation: Permutation Invariant Parking Assortments
- 2023 AAAS Emerging Researchers National Conference in STEM — 9-11 February 2023
  Omni Shoreham Hotel — Washington, District of Columbia
  Poster: Permutation Invariant Parking Functions with cars of assorted lengths
- Joint Mathematics Meetings 2023 — 4-7 January 2023
  John B. Hynes Veterans Memorial Convention Center — Boston, Massachusetts
  Poster: Permutation Invariant Parking Functions with cars of assorted lengths
  Presentation: Permutation Invariant Parking Functions with Cars of Arbitrary Lengths
- Field of Dreams Conference 2022 — 4-6 November 2022
  The Graduate Hotel, University of Minnesota-Twin Cities — Minneapolis, Minnesota
- 2022 SACNAS National Diversity in STEM Conference — 27-29 October 2022
  Pedro Roselló Convention Center — San Juan, Puerto Rico
  Poster: The Study of $\tau_{(n)}$-primes
- 2022 Gulf Coast Undergraduate Research Symposium — 8-9 October 2022
  William Marsh Rice University — Houston, Texas
  Presentation: Properties of $\tau_{(n)}$-primes
- Summer@ICERM 2022: Computational Combinatorics — 3 August 2022
  Institute for Computational and Experimental Research in Mathematics — Providence, Rhode Island
  Presentation: On Permutation-Invariant Parking Sequences
- 2022 ACS Junior Technical Meeting-Puerto Rico Interdisciplinary Scientific Meeting — 9 April 2022
  University of Puerto Rico at Humacao — Humacao, Puerto Rico
  Presentation: The Study of $\tau_{(n)}$-primes
- Joint Mathematics Meetings 2022 — 6-9 April 2022
  Poster: Improving Bounds of Hermitian-Lifted Codes — Virtual Conference
- 37th Interuniversity Mathematical Sciences Research Seminar — 25-26 February 2022
  Poster: The Study of $\tau_{(n)}$-primes — Virtual Conference
  Presentation: Improving Bounds of Hermitian-Lifted Codes
- 2021 Math REU Conference@Clemson University — 19 July 2021
  Presentation: Improved Hermitian-Lifted Codes — Virtual Conference
- 2021 ACS Junior Technical Meeting-Puerto Rico Interdisciplinary Scientific Meeting — 23-24 April 2021
  Presentation: The Study of $\tau_{(n)}$-atoms — Virtual Conference
- 35th Interuniversity Mathematical Sciences Research Seminar — 6-7 March 2020
  University of Puerto Rico at Cayey — Cayey, Puerto Rico
  Poster: The Study of $\tau_{(n)}$-atoms

Academics and Coursework

Purdue University

Coursework (Graduate):

MA 59800: Theory and Applications of Hamilton-Jacobi Equations — June 2026–August 2026
MA 61500: Numerical Methods for Partial Differential Equations — January 2026–May 2026
MA 53200: Elements of Stochastic Processes — January 2026–May 2026
MA 59800ZAT: Hamiltonian Dynamics — January 2026–May 2026

MA 55400: Linear Algebra	August 2025–December 2025
MA 51900: Introduction to Probability	August 2025–December 2025
MA 57300: Numerical Solutions of Ordinary Differential Equations	August 2025–December 2025
MA 59500OT: Computational Optimal Transport and Deep Generative Models	January 2025–May 2025
MA 59800ZY: Topics in Dynamical Systems (Bifurcation Theory)	January 2025–May 2025
MA 54600: Introduction to Functional Analysis	January 2025–May 2025
MA 59500AFF: Analytic Theory of Function Fields	August 2024–December 2024
MA 59500MM: Introduction to Mathematical Modeling	August 2024–December 2024
MA 54300: Introduction to Ordinary Differential Equations and Dynamical Systems	January 2024–May 2024
MA 54400: Real Analysis and Measure Theory	January 2024–May 2024
MA 55300: Introduction to Abstract Algebra	August 2023–December 2023
MA 53000: Functions of a Complex Variable I	August 2023–December 2023

University of Puerto Rico, Mayagüez Campus

Coursework (Graduate):

MATE 6101: Number Theory I	August 2022–December 2022
MATE 5150: Linear Algebra	January 2021–May 2021

Coursework (Undergraduate):

MATE 4000: Elements of Topology	August 2022–December 2022
MATE 4052: Advanced Calculus II	January 2022–May 2022
MATE 4051: Advanced Calculus I	August 2021–December 2021
MATE 4061: Numerical Analysis I	August 2021–December 2021
MATE 4021: Introduction to Mathematical Logic	August 2021–December 2021
MATE 4010: Introduction to Complex Variables	January 2021–May 2021
ESMA 4001: Mathematical Statistics I	January 2021–May 2021
MATE 4020: Partial Differential Equations and Fourier Series	August 2020–December 2020
MATE 4008: Introduction to Algebraic Structures	August 2020–December 2020
MATE 3040: Theory of Numbers	August 2020–December 2020
MATE 4009: Ordinary Differential Equations	June 2020–July 2020
MATE 4031: Introduction to Linear Algebra	January 2020–May 2020
COMP 3010: Introduction to Computer Programming I	January 2020–May 2020
MATE 3063: Calculus III	January 2020–May 2020
MATE 3032: Calculus II	August 2019–December 2019
MATE 3020: Introduction to Mathematical Foundations	August 2019–December 2019
MATE 3031: Calculus I	June 2019–July 2019
MATE 3172: Precalculus II	January 2019–May 2019
MATE 3171: Precalculus I	August 2018–December 2018